CUSTOMER CHURN ANALYSIS

Exploratory Data Analysis Report

This report examines customer churn in a telecommunications dataset, comparing churn rates across contract type, payment method, customer tenure, internet service, and senior-citizen status. All percentages in this version are calculated directly from the supplied CSV dataset.

26.54%	7,043	1,869
Overall churn rate	Customers in dataset	Customers who churned

Executive Summary

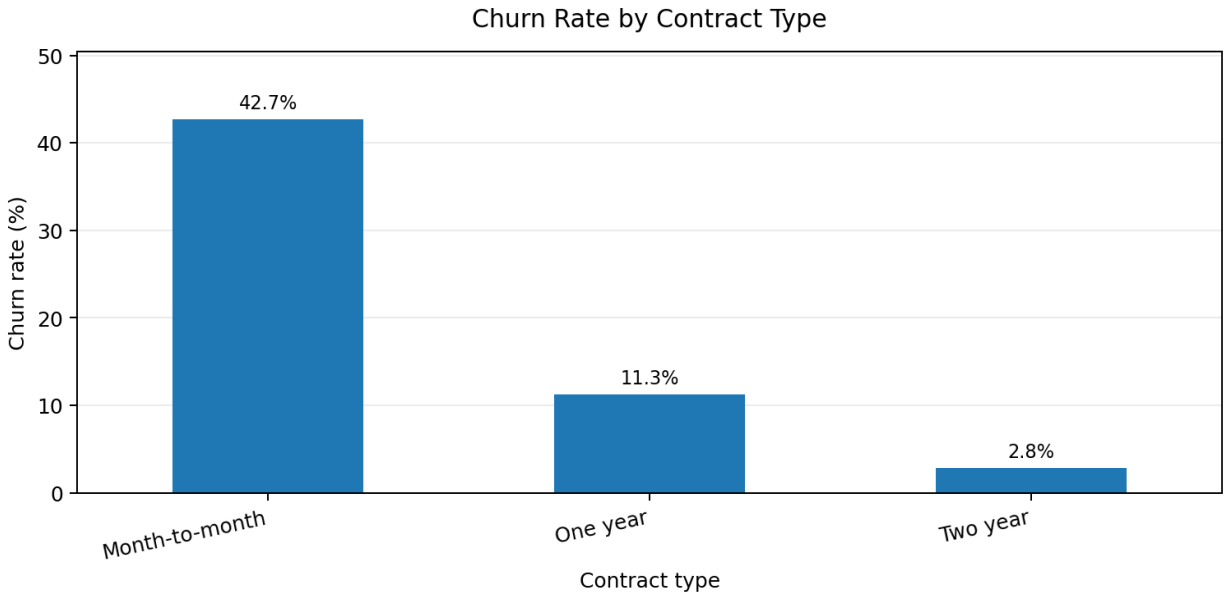
The dataset contains 7,043 customer records, of which 1,869 are marked as churned (26.54%). Churn is highest among customers on month-to-month contracts (42.71%), customers paying by electronic check (45.29%), and customers with less than one year of tenure (48.28%). These are associations in the dataset, not proof that any one factor causes churn.

Verified Key Figures

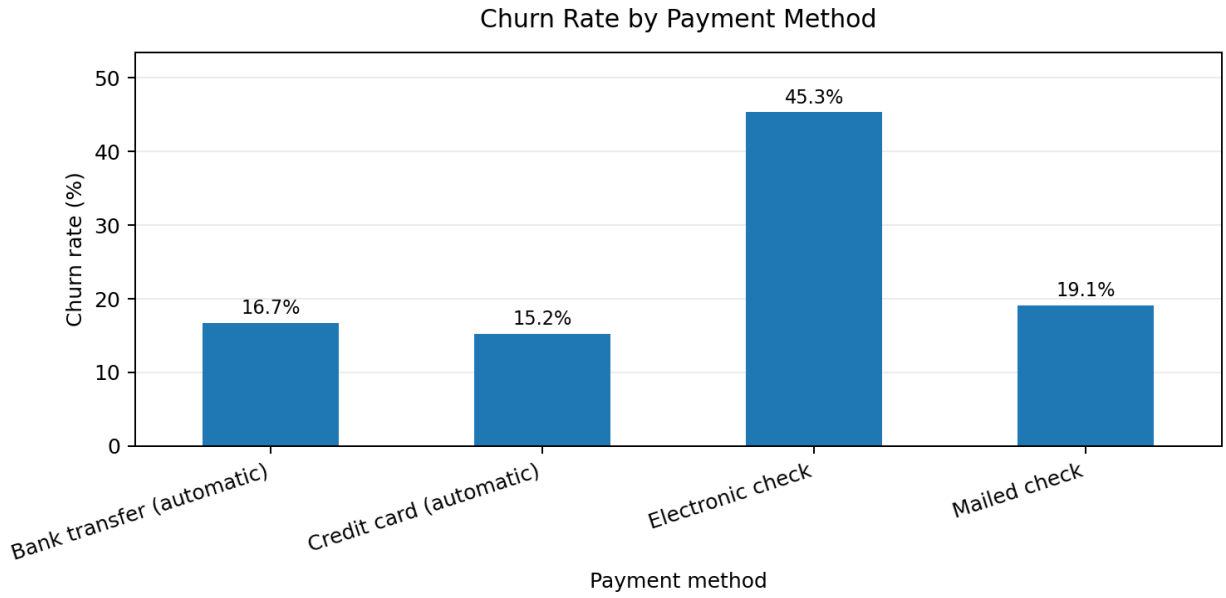
Factor	Group	Churn rate
Contract type	Month-to-month	42.71%
Contract type	One year	11.27%
Contract type	Two year	2.83%
Payment method	Electronic check	45.29%
Customer tenure	Under 1 year (0–11 months)	48.28%
Customer tenure	1–3 years (12–36 months)	25.95%
Customer tenure	Over 3 years (37+ months)	11.93%
Internet service	Fiber optic	41.89%
Internet service	DSL	18.96%
Customer segment	Senior citizen	41.68%
Customer segment	Non-senior citizen	23.61%

Data quality note: 11 rows have blank TotalCharges values. These blanks were treated as missing for this report; they do not affect the churn-rate calculations above because those calculations use the Churn column.

1. Contract Type and Payment Method

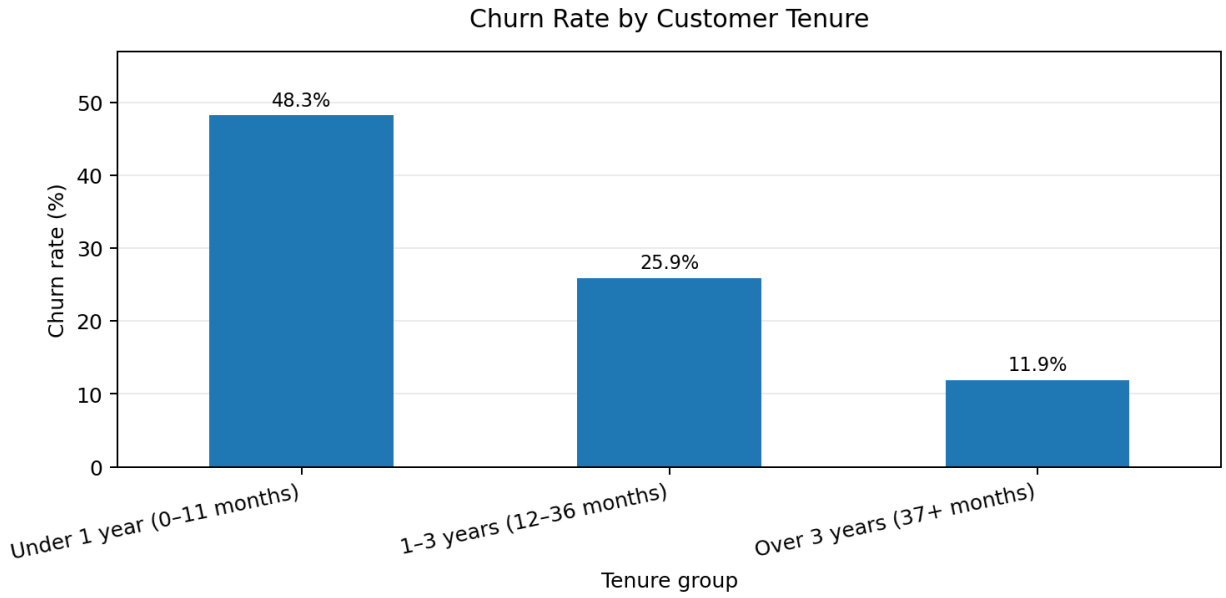


Month-to-month customers have a churn rate of 42.71%, substantially higher than one-year contracts (11.27%) and two-year contracts (2.83%). This makes contract type a useful area to investigate when designing retention initiatives.

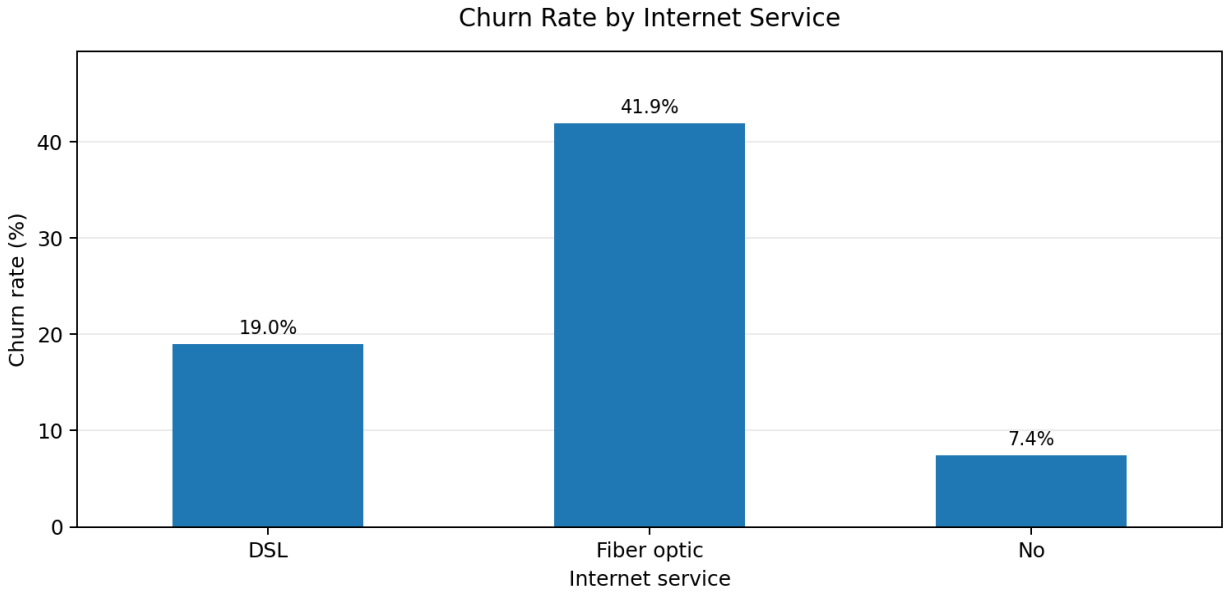


Electronic-check users have a churn rate of 45.29%, higher than the other listed payment methods. The dataset shows an association; further investigation would be needed to understand why this group churns more often.

2. Tenure, Internet Service, and Customer Segment

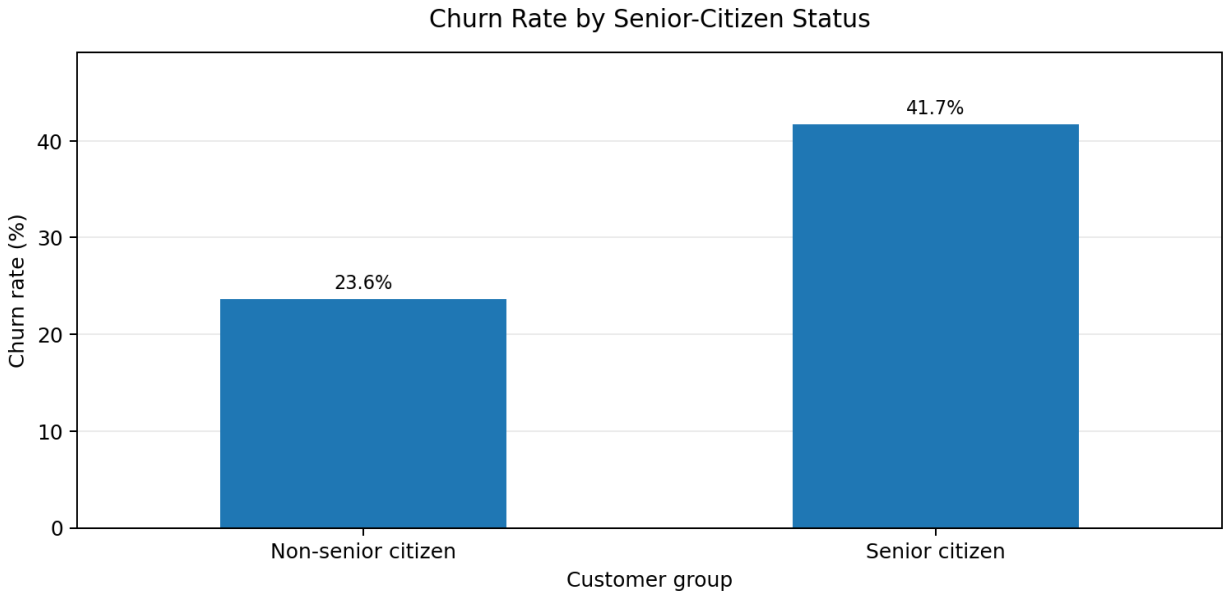


Tenure bands are defined as under 1 year (0–11 months), 1–3 years (12–36 months), and over 3 years (37+ months). Churn falls from 48.28% in the under-one-year group to 25.95% in the 1–3 year group and 11.93% among customers with more than three years of tenure. Early onboarding and issue resolution may deserve attention.



Fiber-optic customers have a churn rate of 41.89%, compared with 18.96% for DSL and 7.40% for customers with no internet service. These differences warrant further investigation into pricing, service quality, customer expectations, and support experience.

3. Senior-Citizen Status, Recommendations, and Conclusion



The churn rate is 41.68% for senior citizens and 23.61% for non-senior citizens. This difference may support further investigation into whether support preferences, plan suitability, or other factors vary by customer segment. The result should not be interpreted as a causal explanation.

Recommended next steps

• Test transparent plan options or incentives for customers on month-to-month contracts.

• Review payment-related support issues and offer clear, optional payment alternatives.

• Strengthen onboarding and early customer check-ins during the first year.

• Investigate fiber-optic customer experience using additional service-quality, pricing, and support data.

• Evaluate customer-support needs across segments inclusively and avoid assuming that a group label explains individual behaviour.

Conclusion

The supplied dataset shows that churn varies across contract type, payment method, tenure, internet service, and senior-citizen status. The most pronounced differences in this analysis appear for month-to-month contracts, electronic-check payment, and shorter tenure. These findings can guide further analysis and retention experiments, but additional evidence is needed before drawing conclusions about causes.

Method: churn rate is calculated as the number of records with Churn = Yes divided by all records in that group. Rates are rounded to two decimal places. Tenure bands are defined explicitly above. Dataset: Customer Churn.csv (7,043 records).